

LLM Engineering Final Curriculum 2026 - 8 phase roadmap for self-learners

LLM Engineering — Final Curriculum 2026

ML/DL Complete → Ab Ye Seekho

Stanford CS224G 2026 · Reddit r/learnmachinelearning · Towards Data Science · Second Talent Hiring Data

Tumhare liye special note: ML/DL already complete hai — Phase 01 fast hoga. Realistic timeline: **12–14 weeks** (general 17–19 hai). Aaj se Phase 01 shuru karo. FastAPI docs kholo, ek endpoint banao.

PHASE 01 Engineering Foundations

1–2 Weeks

Backend solid karo — baad mein ye sab assume kiya jayega

- Python Advanced: Async/Await, Decorators, Generators (FastAPI ke liye critical)
- Pydantic v2: Data validation, Type hinting — LLM structured outputs ka base
- FastAPI: Clean architecture, Dependency Injection, Router splitting
- Docker + Docker Compose: Multi-container setup (App + VectorDB + DB)
- Git/GitHub: Branching, PRs, portfolio-ready commits

YouTube search terms

- "FastAPI full course python 2025"
- "Docker compose tutorial beginner 2025"
- "Pydantic v2 tutorial python"

PHASE 02 LLM Core & Reasoning Models

2 Weeks

LLMs se programmatically baat karna — Stanford CS224G Week 2 aligned

- Tokens, Embeddings, Context Window — aur in ko manage karna
- Reasoning Models: o1/o3/DeepSeek R1/Claude — kab use karein kab nahi
- Prompting: Zero-shot, Few-shot, Chain-of-Thought, ReAct pattern
- APIs: OpenAI, Anthropic Claude, Gemini, Groq (ultra-fast inference)
- Structured Outputs: Pydantic + Instructor library — secure JSON responses
- Function/Tool Calling: LLM ko Python functions execute karwana
- Cost & Latency tradeoffs — production mein roz ka kaam

Warning: Transformer math mein mat doob jao — yahan self-taught log QUIT karte hain. Concept samajhna kaafi hai.

YouTube search terms

- "OpenAI structured outputs pydantic 2025"
- "Chain of thought prompting tutorial"
- "Instructor library python LLM"

→ "reasoning models vs standard models explained"

PHASE 03 Context Engineering & RAG 2.0

3 Weeks

Clients ka private data use karke enterprise-grade search banana

- Context Engineering: System prompts, token budget, memory systems
- RAG full pipeline: Chunking → Embedding → Retrieval → Generation
- Chunking Strategies: Fixed, Semantic, Hierarchical
- Vector DBs: ChromaDB (local), Pinecone / Qdrant (production scale)
- Hybrid Search: Dense Embeddings + Sparse BM25, Reciprocal Rank Fusion
- Reranking: Cohere Rerank / Cross-Encoder — context quality drastically improve
- RAGAS Framework: Retrieval aur generation accuracy evaluate karna
- LLM Caching: GPTCache — client API bills 40% tak reduce

Warning: 75% log basic RAG banate hain aur stuck ho jaate hain. Hybrid search + reranking skip karna = poor production results.

YouTube search terms

- "RAG pipeline python tutorial 2025"
- "ChromaDB fastapi integration"
- "hybrid search BM25 langchain"
- "Cohere rerank tutorial python"
- "RAGAS evaluation RAG pipeline"

PHASE 04 Fine-Tuning — LoRA / QLoRA

2 Weeks

Ye skill jo \$150–\$250/hr freelance rate deti hai

- LoRA: Efficiently fine-tune bina full model train kiye
- QLoRA: 4-bit quantization + LoRA — consumer GPU ya free Colab pe bhi
- Dataset Preparation: Instruction tuning format — quality > quantity
- Hugging Face PEFT library: Production fine-tuning ka standard
- Unsloth: 2x faster training — free Google Colab pe chalega
- DPO (Direct Preference Optimization): RLHF se simple alignment
- Evaluation: LLM-as-Judge — fine-tuned model test karna

Ye woh skill hai jo tujhe wrapper developer se utha kar real LLM Engineer banati hai. Clients 2x pay karte hain fine-tuning ke liye.

YouTube search terms

- "QLoRA fine tuning llama python 2025"
- "Unsloth fine tuning tutorial colab"
- "PEFT LoRA hugging face tutorial"
- "DPO fine tuning tutorial 2025"

PHASE 05 Agentic AI & Multi-Agent Systems

3 Weeks

Autonomous systems banana jo complex business workflows handle karein

- LangChain Basics: Chains, Memory — sirf ek framework pakro pehle
- LangGraph (Critical): Stateful, cyclical multi-agent workflows — 2026 industry standard
- MCP (Model Context Protocol): 2026 ka naya standard — tools ko LLMs se connect karna
- CrewAI: Multi-agent roles, goals, task orchestration rapid development ke liye
- Human-in-the-Loop (HITL): User approval steps agentic workflows mein inject karna
- Voice AI (Bonus): Realtime voice agents — Stanford CS224G mein explicitly cover

Warning: LangChain + LlamaIndex + Haystack ek saath mat seekho — Tutorial Hell mein phase jaoge. Sirf LangGraph pakro pehle.

YouTube search terms

- "LangGraph crash course 2025"
- "MCP model context protocol tutorial 2025"
- "CrewAI multi agent fastapi python"
- "LangGraph human in the loop"

PHASE 06 Full-Stack AI Packaging

1–2 Weeks

AI backend ko complete web app banana — clients ko dikhane ke liye

- FastAPI Auth: JWT authentication, session-wise chat history tracking
- Database: Supabase (PostgreSQL) ya MongoDB — chat logs + user data
- Chainlit ya Streamlit: 1 din mein ChatGPT jaisa premium UI ready
- AI Safety & Guardrails: Prompt injection defense, NeMo Guardrails
- Red Teaming: Edge cases pe model test karna — production must

YouTube search terms

- "Chainlit crash course chatbot UI python"
- "Streamlit chatbot with history 2025"
- "NeMo guardrails LLM safety tutorial"

PHASE 07 LLMops & Production Deployment

2 Weeks

AI systems ko scale, monitor aur cloud par deploy karna

- Ollama: Local open-source models test karna (Llama 3, Mistral)
- vLLM / TGI: High-throughput production serving — enterprise clients yahi maangte hain
- LangSmith / Phoenix: Har prompt run, latency, token costs track karna
- Cloud Deploy: Railway, Render, ya AWS Lightsail — Docker containers ke sath

YouTube search terms

- "vLLM production serving tutorial 2025"
- "LangSmith monitoring LLM tutorial"
- "deploy fastapi docker railway 2025"

Real clients ke liye working products — code nahi, chalta hua Loom demo

Project 1 — Enterprise RAG Platform
Multi-file upload (PDFs + Websites), persistent chat history, hybrid search, RAGAS evaluation dashboard.

Project 2 — Fine-Tuned Domain Expert
Custom LoRA model on niche dataset (legal/medical/Urdu). Deploy via FastAPI + Ollama. Wrapper developers ye nahi kar sakte.

Project 3 — Multi-Agent Cold Outreacher
CrewAI + LangGraph: Website scrape → lead research → personalized cold email. Real business value immediately.

Project 4 — Customer Support Agent with HITL
LangGraph agent jo SQL reads/writes kare, billing handle kare, high-value actions pe manager approval maange.

Golden rules

- Rule of One

Ek waqt mein sirf ek framework. LangGraph shuru kiya? LlamaIndex ko 2 hafte touch mat karo.
- No Tutorial Hell

30 min video dekhi? Agle 2 ghante VS Code mein khud likho aur errors debug karo.
- Build to Sell

Har project ki Loom screen recording. Clients code nahi, chalta demo dekhte hain.
- Date Filter

YouTube pe views nahi — Upload Date filter. Sirf last 6-12 months ki videos dekho.
- Project > Courses

Jo log projects banate rehte hain woh hamesha pehle hire hote hain course collectors se.
- Fine-Tune to Stand Out

RAG sab karte hain. Fine-tuning jo karta hai woh 2x earn karta hai. Skip mat karo.

Timeline & earning progression

Phase	Topic	Duration	Earning unlock
01	Engineering Foundations	1-2w	—
02	LLM Core + Reasoning Models	2w	Basic API projects
03	Context Engineering + RAG 2.0	3w	\$50–80/hr
04	Fine-Tuning LoRA/QLoRA	2w	\$100–150/hr
05	Agentic AI + MCP	3w	\$150–200/hr
06	Full-Stack AI Packaging	1-2w	Complete products
07	LLMOps + Production	2w	\$200–250/hr
08	Portfolio — 4 Projects	3w	Job-ready

Total: 12–14 weeks* → Production LLM Engineer

* ML/DL already complete hone ki wajah se tumhare liye 12-14 weeks — general 17-19 hai

Sources: Stanford CS224G Winter 2026 · Towards Data Science · DEV Community · Reddit r/learnmachinelearning · Second Talent Hiring Data April 2026